Air Quality

Pupil sheet

Name: _____ **Date:** _____

Safety: Do not move the boxes unless told to; the teacher opens the window; never climb on the sill.

1. What I see on the screen

The screen shows **one gauge per box** and a summary line: Δ**CO₂**, the difference between the highest box and the lowest. With only one box it reads **Connect at least two boxes**. Each box has its own name and colour, the same on the gauge and on the chart.

Under the Δ line the app writes two lines. The **Level** comes from the highest value: from 1200 ppm "the air is getting stale", from 2000 ppm "open the windows now". The **Difference** uses *fixed thresholds* — 300 and 600 ppm — which know nothing about our boxes. **I check it against my own threshold.**

Our detection threshold (from co-location) = _____ **ppm**
A difference smaller than that *cannot be told* from noise.

2. What I do on my own

1. Co-location. Every box on the same table, ten minutes. I record ΔCO_2 over the last five minutes; the largest value is the threshold above.

Minute	6	7	8	9	10
Δ (ppm)					

First I predict. The corner furthest from the window will be about _____ ppm higher than the spot beside it.

because _____

_____

2. Every location, at the same moment. Time: _____ How many of us: _____ Window: ☐ shut ☐ open

Location	Box	CO_2 (ppm)	Band (level)
Far corner			
By the window			
By the door			
Middle of the room			

3. I compare every pair with my threshold. "Borderline" is a correct answer, not a dodge.

The pair	Δ (ppm)	Verdict (circle one)
Far corner vs window		detected / borderline / not detected
Far corner vs door		detected / borderline / not detected
Middle vs window		detected / borderline / not detected

4. Airing. We open the window. ΔCO_2 immediately afterwards: _____ ppm. Did it rise? ☐ yes ☐ no
After ten minutes: _____ ppm.

3. The conclusion I write

I predicted that, and the data show that _____

_____

When we opened the window, Δ rose because _____

_____

The location most in need of air is _____, because _____

_____

Exit ticket

What is our detection threshold, and how do we know it?

_____